Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A particle executes SHM with an angular frequency $\omega = 4\pi \text{ rad/s}$. If it is at its extreme position initially, then find the instants when it is at a distance $\sqrt{3}/2$ times its amplitude, from the mean position.

(A) $\frac{13}{24}$ s

(B) $\frac{13}{25}$ s

(C) $\frac{13}{4}$ s

(D) $\frac{3}{24}$ s

Q.2

A coin is placed on a horizontal platform which undergoes vertical simple harmonic motion of angular frequency ω. The amplitude of oscillation is gradually increased. The coin will leave contact with the platform for the first time

(A) at the mean position of the platform

(B) for an amplitude of $\frac{g}{\omega^2}$

(C) for an amplitude of $\frac{g^2}{\omega^2}$

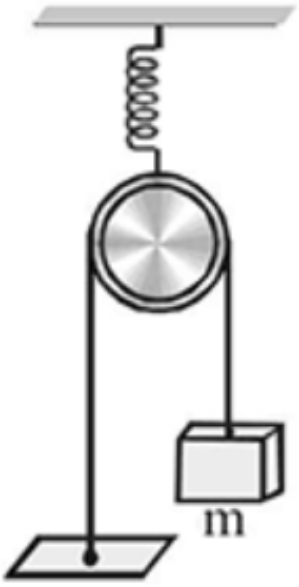
(D) at the highest position of the platform

Q.3

In the figure shown, the spring is light and has a force constant k. The pulley is light and smooth and the string is light. The suspended block has a mass m. On giving a slight displacement vertically to the block in the downward direction from its equilibrium position the block executes S.H.M. on being released with time period

@IITJEE_Advanced

T. Then



(A) $\mathrm{T}=2\pi\sqrt{\frac{\mathrm{m}}{\mathrm{k}}}$

(B) $\mathrm{T}=2\pi\sqrt{\frac{\mathrm{m}}{2\mathrm{k}}}$

(C) $\mathrm{T}=2\pi\sqrt{\frac{2\,\mathrm{m}}{\mathrm{k}}}$

(D) $\mathrm{T}=4\pi\sqrt{\frac{\mathrm{m}}{\mathrm{k}}}$

Q.4

A point particle of mass $0.1\ \mathrm{kg}$ is executing S.H.M. of amplitude $0.1\ \mathrm{m}$. When the particle passes through the mean position, its kinetic energy is 8×10^{-3} joule. Obtain the equation of motion of this particle, if the initial phase of oscillation is 45°.

(A) $y=0.1\sin\left(\pm4t+\frac{\pi}{4}\right)$

(B) $\mathrm{y}=0.2\sin\left(\pm4\mathrm{t}+\frac{\pi}{4}\right)$

(C) $\mathrm{y}=0.1\sin\left(\pm2\mathrm{t}+\frac{\pi}{4}\right)$

(D) $\mathrm{y}=0.2\sin\left(\pm2\mathrm{t}+\frac{\pi}{4}\right)$

Q.5

The displacement of an object attached to spring and executing simple harmonic motion is given by $x=2\times10^{-2}\cos\pi\mathrm{t}$ metre. The time at which the maximum speed first occurs is

(A) $0.25\ \mathrm{s}$

(B) $0.5\ \mathrm{s}$

(C) $0.75\ \mathrm{s}$

(D) $0.125\ \mathrm{s}$

@IITJEE_Advanced

Q.6

A point mass oscillates along the x -axis according to the law $x = x_0 \cos(\omega t - \pi/4)$. If the acceleration of the particle is written as $a = A\cos(\omega t - \delta)$, then

(A) $A = x_0\omega^2, \delta = 3\pi/4$

(B) $A = x_0, \delta = -\pi/4$

(C) $A = x_0\omega^2, \delta = \pi/4$

(D) $A = x_0\omega^2, \delta = -\pi/4$

Q.7

A particle of mass m is attached to a spring (of spring constant k) and has a natural angular frequency ω_0. An external force $F(t)$ proportional to $\cos \omega t$ $(\omega \neq \omega_0)$ is applied to the oscillator. The displacement of the oscillator will be proportional to

(A) $\frac{1}{m\left(\omega_0^2+\omega^2\right)}$

(B) $\frac{1}{m\left(\omega_0^2-\omega^2\right)}$

(C) $\frac{m}{\omega_0^2-\omega^2}$

(D) $\frac{m}{\left(\omega_0^2+\omega^2\right)}$

Q.8

An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M. The piston and the cylinder have equal cross sectional area A. When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0. The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency

(A) $\frac{1}{2\pi}\frac{\mathrm{A}\gamma\mathrm{P}_0}{\mathrm{V}_0\mathrm{M}}$

(B) $\frac{1}{2\pi}\frac{V_0\mathrm{M}\mathrm{P}_0}{\mathrm{A}^2\gamma}$

(C) $\frac{1}{2\pi}\sqrt{\frac{\mathrm{A}^2\gamma\mathrm{P}_0}{\mathrm{M}\mathrm{V}_0}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{\mathrm{M}\mathrm{V}_0}{\mathrm{A}\gamma\mathrm{P}_0}}$

Q.9

The amplitude of a damped oscillator becomes $\left(\frac{1}{3}\right)^{\mathrm{rd}}$ in 2
seconds. If its amplitude after 6 seconds is $\frac{1}{\mathrm{n}}$ times the original amplitude, the value of n is

(A) 25

@IITJEE_Advanced

(B) 29

(C) 23

(D) 27

Q.10

The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10 s it will decrease to α times its original magnitude, then find the value of α ?

(A) 0.792

(B) 0.739

(C) 0.793

(D) 0.729

Q.11

A particle performs SHM in a straight line. In the first second, starting from rest, it travels a distance a and in the next second it travels a distance b in the same direction. The amplitude of the SHM is

(A) $a-b$

(B) $\frac{2a-b}{3}$

(C) $\frac{2a^2}{3a-b}$

(D) None of these

Q.12

A simple harmonic oscillator of angular frequency $2\,\mathrm{rad\,s^{-1}}$ is acted upon by an external force $\mathrm{F}=\sin t$. If the oscillator is at rest in its equilibrium position at $t=0$, its position at later times is given by:

(A) $\sin t+\frac{1}{2}\cos 2t$

(B) $\cos t-\frac{1}{2}\sin 2t$

(C) $\sin t-\frac{1}{2}\sin 2t$

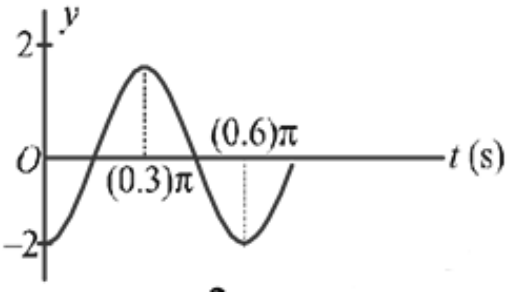
(D) $\sin t+\frac{1}{2}\sin 2t$

Q.13

Part of a simple harmonic motion is graphed in the figure, where y is the displacement from the mean position.

@IITJEE_Advanced

The correct equation describing this



(A) $y = 4\cos(0.6t)$

(B) $y = 2\sin\left(\frac{10}{3}t - \frac{\pi}{2}\right)$

(C) $y = 4\sin\left(\frac{10}{3}t + \frac{\pi}{2}\right)$

(D) $y = 2\cos\left(\frac{10}{3}t + \frac{\pi}{2}\right)$

Q.14

The phase difference between the instantaneous velocity and acceleration of a particle executing simple harmonic motion is

(A) π

(B) 0.707π

(C) zero

(D) 0.5π

Q.15

A particle of mass $1\ \text{kg}$ is moving in S.H.M. with an amplitude 0.02 and a frequency of $60\ \text{Hz}$. The maximum force acting on the particle is

(A) $144\pi^2$

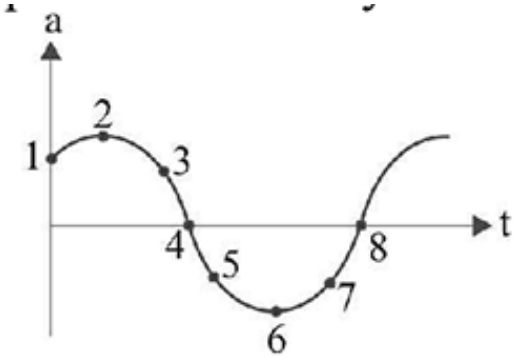
(B) $188\pi^2$

(C) $288\pi^2$

(D) None of these

Q.16

The acceleration of a particle undergoing SHM is graphed in figure. At point 2 the velocity of the particle is



@IITJEE_Advanced

(A) zero

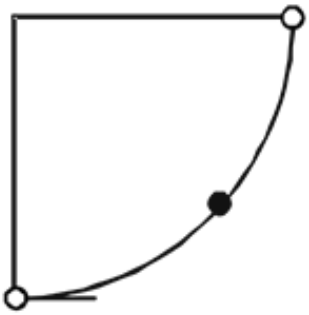
(B) negative

(C) positive

(D) None of these

Q.17

A particle starts with S.H.M. from the mean position as shown in figure below. Its amplitude is A and its time period is T. At one time, its speed is half that of the maximum speed. What is the displacement at that time?



(A) $\frac{\sqrt{2}\,\mathrm{A}}{3}$

(B) $\frac{\sqrt{3}\,\mathrm{A}}{2}$

(C) $\frac{2\,\mathrm{A}}{\sqrt{3}}$

(D) $\frac{3\,\mathrm{A}}{\sqrt{2}}$

Q.18

A body executing linear simple harmonic motion has a velocity of $3\ \mathrm{m/s}$ when its displacement is 4 cm and a velocity of $4\ \mathrm{m/s}$ when its displacement is $3\ \mathrm{cm}$. What is the amplitude of oscillation?

(A) $5\ \mathrm{cm}$

(B) $7.5\ \mathrm{cm}$

(C) $10\ \mathrm{cm}$

(D) $12.5\ \mathrm{cm}$

Q.19

The displacement of a particle in SHM is $x = 10 \sin\left(2t - \frac{\pi}{6}\right)$ metre. When its displacement is $6\ \mathrm{m}$, the velocity of the particle (in ms^{-1}) is

(A) 8

(B) 24

(C) 16

(D) 10

@IITJEE_Advanced

Q.20

Two particles are executing simple harmonic motion of the same amplitude A and frequency ω along the x - axis. Their mean position is separated by distance $X_0\ (X_0 > A)$. If the maximum separation between them is $(\mathrm{X}_0 + \mathrm{A})$, the phase difference between their motion is

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{2}$

Answers & Solutions

Q.1 Answer:

$\frac{13}{24}$ s

Solution:

Let a be the amplitude and ϕ the phase constant for the oscillation.

The displacement equation can be written as

$x = a\sin(4\pi t + \phi)\ \left[\because \omega = 4\pi_{\mathrm{rad/s}}\right]$

Since at $t = 0;\quad x = +a$**, therefore** $4\pi(0) + \phi = \frac{\pi}{2} \Rightarrow \phi = \frac{\pi}{2}$

$\therefore \quad x = a\sin(4\pi t + \pi/2) = a\cos(4\pi t)$

Now let t_n be the instant when the particle's displacement s

$\sqrt{3}a/2, n\mathrm{th}$

Evidently $\pm\frac{\sqrt{3}a}{2} = a\cos(4\pi t_n)$

$\Rightarrow \quad 4\pi t_n = n\pi \pm \frac{\pi}{6} \Rightarrow t_n = \frac{n}{4} \pm \frac{1}{24}$

Taking only the positive values for 't',

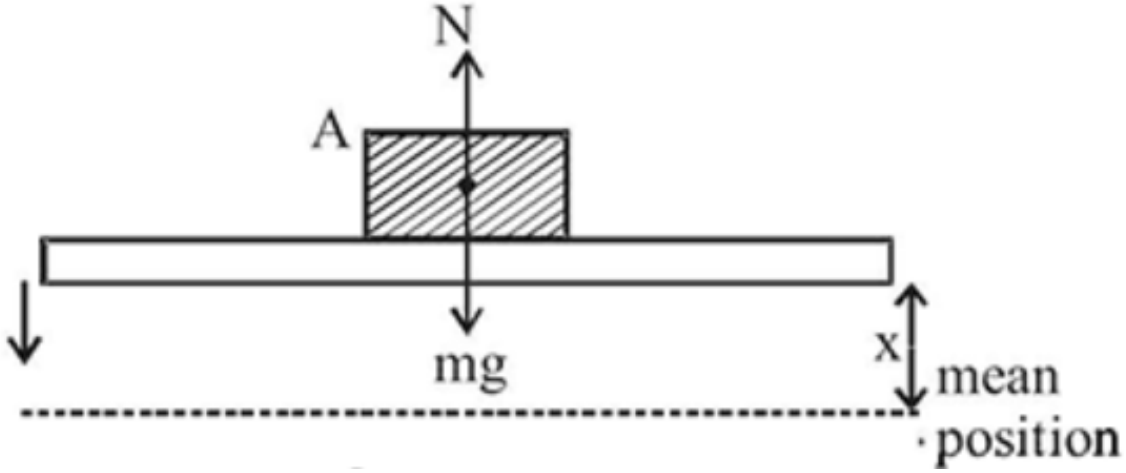
$t_1 = \frac{1}{24}\ \mathrm{s}, t_2 = \frac{5}{24}\ \mathrm{s}, t_3 = \frac{7}{24}\ \mathrm{s}; t_4 = \frac{11}{24}\ \mathrm{s}, t_5 = \frac{13}{24}\ \mathrm{s},\ \text{etc.}$

Q.2 Answer:

for an amplitude of $\frac{g}{\omega^2}$

Solution:

For block A to move in S.H.M.



@IITJEE_Advanced

$mg - N = m\omega^2 x$

where x is the distance from mean position For block to leave contact $\mathrm{N} = 0$

$\Rightarrow \mathrm{mg} = \mathrm{m}\omega^2\mathrm{x} \Rightarrow \mathrm{x} = \frac{\mathrm{g}}{\omega^2}$

Q.3 Answer:

$\mathrm{T} = 4\pi\sqrt{\frac{\mathrm{m}}{\mathrm{k}}}$

Solution:

Let the extension in the spring be x_0 **at equilibrium. If** F_0 **be the tension in the string then** $\mathrm{F}_0 = \mathrm{kx}_0$**. Further if** T_0 **is the tension in the thread then** $\mathrm{T}_0 = \mathrm{mg}$ **and** $2T_0 = \mathrm{kx}_0$

Let the mass m **be displaced through a slight displacement** x **downwards. Let the the new tension in the string and spring be** T **and** F **respectively.** $\Rightarrow \mathrm{F} = \left(\mathrm{x}_0 \frac{\mathrm{x}}{2}\right)$ **and** $\mathrm{F} = 2\ \mathrm{T}$

$\Rightarrow 2\ \mathrm{T} = \mathrm{k}\left(\mathrm{x}_0 \frac{\mathrm{x}}{2}\right)$

$\Rightarrow \quad (\mathrm{T} - \mathrm{T}_0) = \frac{\mathrm{kx}}{4}$

$\Rightarrow$ **Time period** $= 2\pi\sqrt{\frac{\mathrm{m}}{\frac{\mathrm{k}}{4}}} = 4\pi\sqrt{\frac{\mathrm{m}}{\mathrm{k}}}$

Q.4 Answer:

$y = 0.1 \sin\left(\pm 4t + \frac{\pi}{4}\right)$

Solution:

The displacement of a particle in S.H.M. is given by $\mathrm{y} = \mathrm{a}\sin(\omega\mathrm{t} + \phi)$

velocity $= \frac{\mathrm{dy}}{\mathrm{dt}} = \omega\mathrm{a}\cos(\omega\mathrm{t} + \phi)$

The velocity is maximum when the particle passes through the mean position i.e.,

$$\left(\frac{\mathrm{dy}}{\mathrm{dt}}\right)_{\mathrm{max}} = \omega$$

The kinetic energy at this instant is given by

$$\frac{1}{2}\,\mathrm{m}\left(\frac{\mathrm{dy}}{\mathrm{dt}}\right)^2_{\mathrm{max}} = \frac{1}{2}\,\mathrm{m}\omega^2\mathrm{a}^2 = 8 \times 10^{-3}\ \text{joule}$$

or $\frac{1}{2} \times (0.1)\omega^2 \times (0.1)^2 = 8 \times 10^{-3}$

Solving we get $\omega = \pm 4$ **Substituting the values of** a, ω **and** ϕ **in the equation of S.H.M., we get** $\mathrm{y} = 0.1\sin(\pm 4\mathrm{t} + \pi/4)$ **metre**

Q.5 Answer:

$0.5\ \mathrm{s}$

Solution:

@IITJEE_Advanced

Given, displacement $\mathrm{x} = 2 \times 10^{-2} \cos \pi \mathrm{t}$

The magnitude of velocity, $\mathrm{v} = \left|\frac{\mathrm{dx}}{\mathrm{dt}}\right| = 2\pi \times 10^{-2} \sin \pi \mathrm{t}$

So, the velocity will be maximum when $\sin \pi \mathrm{t} = 1$ **where** $\pi \mathrm{t} = \frac{\pi}{2}, \frac{3\pi}{2}, ..$

So for first maximum, $\pi \mathrm{t} = \frac{\pi}{2}$ **or** $\mathrm{t} = 0.5 \text{ s}$

Q.6 Answer:

$A = x_0\omega^2, \delta = 3\pi/4$

Solution:

Here,

$x = x_0 \cos(\omega t - \pi/4)$

$\therefore$ **Velocity,** $v = \frac{dx}{dt} = -x_0\omega \sin\left(\omega t - \frac{\pi}{4}\right)$

Acceleration,

$a = \frac{dv}{dt} = -x_0\omega^2 \cos\left(\omega t - \frac{\pi}{4}\right)$

$= x_0\omega^2 \cos\left[\pi + \left(\omega t - \frac{\pi}{4}\right)\right]$

$= x_0\omega^2 \cos\left(\omega t + \frac{3\pi}{4}\right)$

Acceleration, $a = A\cos(\omega t + \delta) \ldots (2)$

Comparing the two equations, we get $A = x_0\omega^2$ **and** $\delta = \frac{3\pi}{4}$.

Q.7 Answer:

$\frac{1}{m\left(\omega_0^2 - \omega^2\right)}$

Solution:

Equation of displacement is given by $x = A\sin(\omega t + \phi)$

where $A = \frac{F_0}{m\sqrt{\left(\omega_0^2 - \omega^2\right)^2}} = \frac{F_0}{m\left(\omega_0^2 - \omega^2\right)}$

Here damping effect is considered to be zero

$\therefore x \propto \frac{1}{m\left(\omega_0^2 - \omega^2\right)}$

Q.8 Answer:

$\frac{1}{2\pi}\sqrt{\frac{\mathrm{A}^2\gamma \mathrm{P}_0}{\mathrm{MV}_0}}$

Solution:

$\frac{Mg}{A} = P_0$

Let piston is displaced by distance x $Mg - \left(\frac{P_0 x_0^\gamma}{(x_0 - x)^\gamma}\right) A = F_{\text{restoring}}$

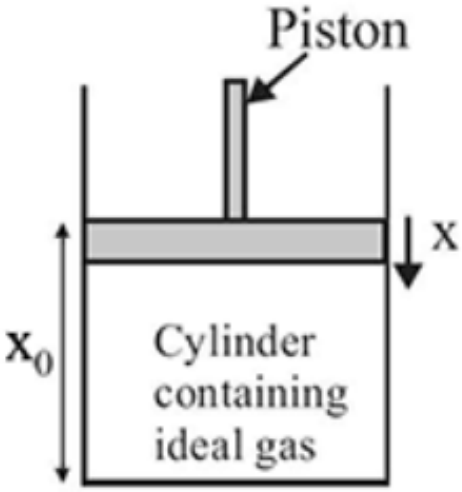
$P_0 A\left(1 - \frac{x_0^\gamma}{(x_0 - x)^\gamma}\right) = F_{\text{restoring}} \quad [x_0 \gg x]$

$F = -\frac{\gamma P_0 A x}{x_0}$

$\therefore$ **Frequency with which piston executes SHM.**

@IITJEE_Advanced

$f = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0 A}{x_0 M}} = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0 A^2}{M V_0}}$



Q.9 Answer:

27

Solution:

Amplitude of a damped oscillator at any instant t is given by $\mathrm{A} = \mathrm{A}_0 \mathrm{e}^{-\mathrm{bt}/2\,\mathrm{m}}$

where A_0 is the original amplitude From question, When $\mathrm{t} = 2\ \mathrm{s},\ \mathrm{A} = \frac{\mathrm{A}_0}{3}$

$\therefore \frac{\mathrm{A}_0}{3} = \mathrm{A}_0 \mathrm{e}^{-2\,\mathrm{b}/2\,\mathrm{m}}$

or, $\frac{1}{3} = \mathrm{e}^{-\mathrm{b}/\mathrm{m}}$

When $\mathrm{t} = 6\ \mathrm{s},\ \mathrm{A} = \frac{\mathrm{A}_0}{\mathrm{n}}$

$\therefore \frac{\mathrm{A}_0}{\mathrm{n}} = \mathrm{A}_0 \mathrm{e}^{-6\,\mathrm{b}/2\,\mathrm{m}}$

or, $\frac{1}{\mathrm{n}} = \mathrm{e}^{-3\,\mathrm{b}/\mathrm{m}} = \left(\mathrm{e}^{-\mathrm{b}/\mathrm{m}}\right)^3$

or, $\frac{1}{\mathrm{n}} = \left(\frac{1}{3}\right)^3$ **(Using eq. (i))**

$\therefore \mathrm{n} = 3^3 = 27$

Q.10 Answer:

0.729

Solution:

$\cdots A = A_0 e^{-\frac{bt}{2m}}$

(where, A_0 **= maximum amplitude) According to the questions, after 5 second,**

$0.9\ \mathrm{A}_0 = \mathrm{A}_0 \mathrm{e}^{-\frac{\mathrm{b}(5)}{2\,\mathrm{m}}}$ **... (i)**

After 10 more second,

$A = A_0 e^{-\frac{b(15)}{2m}}$ **... (ii)**

From eq^{n}s (i) and (ii) $A = 0.729\ \mathrm{A}_0$

$\therefore \alpha = 0.729$

Q.11 Answer:

$\frac{2a^2}{3a-b}$

Solution:

@IITJEE_Advanced

Assumption:-

$y = A\cos\omega t$

To, $A - a = A\,\text{Cos}\,\omega$ **of** $t = 1$

Now at $t = 2$

$A - (a+b) = A\,\text{Cos}\,2\omega$

$A - a - b = A\left(2\cos^2\omega - 1\right)$

$$A - a - b = A\left[2 \times \left(\frac{A-a}{A}\right)^2 - 1\right]$$

$$\Rightarrow A - a - b = A\left[\frac{2\left(A^2 + a^2 - 2Aa\right)}{A^2} - 1\right]$$

$$\Rightarrow A - a - b = A\left[2\left(\frac{A^2 + a^2 - 2Aa}{A^2}\right) - 1\right]$$

$$\Rightarrow A - a - b = 2\left(\frac{A^2 + a^2 + 2Aa}{A}\right) - A$$

$$\Rightarrow 2A = a + b + 2/A + \frac{20^2}{A} \cdots - 4a$$

$$\Rightarrow \frac{2a^2}{A} = 3a - b \Rightarrow A = \frac{2a^2}{3a-b}$$

$$A = \frac{2a^2}{3a-b}$$

Q.12 Answer:

$\sin t - \frac{1}{2}\sin 2t$

Solution:

As we know, $F = ma \Rightarrow a \propto F$ **or,** $a \propto \sin$

$\Rightarrow \frac{dv}{dt} \propto \sin t \Rightarrow \int_0^0 dV \propto \int_0^t \sin t dt$

$\text{V} \propto -\cos t + 1$

$\int_0^x dx = \int_0^t (-\cos t + 1)dt$

$x = \sin t - \frac{1}{2}\sin 2t$

Q.13 Answer:

$y = 2\sin\left(\frac{10}{3}t - \frac{\pi}{2}\right)$

Solution:

From the figure, $\frac{T}{2} = 0.3\pi, \therefore T = 0.6\pi$ **and** $\phi_0 = -\frac{\pi}{2}$.

Thus $y = A\sin(\omega t + \phi_0) = 2\sin\left(\frac{2\pi}{0.6\pi}t + -\frac{\pi}{2}\right)$

$= 2\sin\left(\frac{10t}{3} - \frac{\pi}{2}\right).$

Q.14 Answer:

@IITJEE_Advanced

0.5π

Solution:

Let $y = A\sin\omega t$

$v_{\text{inst}} = \frac{dy}{dt} = A\omega\cos\omega t = A\omega\sin(\omega t + \pi/2)$

Acceleration $= -A\omega^2\sin\omega t = A\omega^2\sin(\pi + \omega t)$

$\therefore \phi = \frac{\pi}{2} = 0.5\pi$

Q.15 Answer:

$288\pi^2$

Solution:

Max. force $=$ mass $\times$ max . acceleration $= \mathrm{m}4\pi^2\mathrm{v}^2\mathrm{a} = 1 \times 4 \times \pi^2 \times (60)^2 \times 0.02$

$= 288\pi^2$

Q.16 Answer:

zero

Solution:

At point 2, the acceleration of the particle is maximum, which is at the extreme position. At extreme position, the velocity of the particle will be zero.

Q.17 Answer:

$\frac{\sqrt{3}\,\mathrm{A}}{2}$

Solution:

$v = \omega\left[A^2 - x^2\right]^{1/2} \Rightarrow x = \left[A^2 - \frac{v^2}{\omega^2}\right]^{1/2}$

Given that $\mathrm{v} = \frac{\mathrm{v_{max}}}{2} = \frac{\mathrm{A}\omega}{2}$.

so, $\left[A^2 - \frac{A^2\omega^2}{4\omega^2}\right]^{1/2} = \frac{\sqrt{3}}{2}\cdot A$

Q.18 Answer:

5 cm

Solution:

Velocity in SHM is given by $v = \omega\sqrt{a^2 - y^2}$

At $y = 4$ cm $= 0.04$ m, $\mathrm{v} = 3$ m/s

$\therefore 3 = \omega\sqrt{\mathrm{a}^2 - (0.04)^2}$

At $\mathrm{y} = 3$ cm $= 0.03$ m, $\mathrm{v} = 4$ m/s

$\therefore 4 = \omega\sqrt{\mathrm{a}^2 - (0.03)^2}$

Dividing (2) by (1), we get $\mathrm{a} = 0.05 = 5$ cm

Q.19 Answer:

@IITJEE_Advanced

16

Solution:

(Given, $x = 10 \sin\left(2t - \frac{\pi}{6}\right)$
$A = 10$ and $\omega = 2$ Hz
$\therefore v = \omega\sqrt{A^2 - d^2} = 2\sqrt{(10)^2 - (6)^2}$
$= 2\sqrt{100 - 36} = 2 \times 8 = 16 \text{ m s}^{-1}$

Q.20 Answer:

$\frac{\pi}{3}$

Solution:

$x_1 = A \sin(\omega t + \phi_1)$
$x_2 = A \sin(\omega t + \phi_2)$
$x_1 - x_2 = A\left[2 \sin\left[\omega t + \frac{\phi_1 + \phi_2}{2}\right] \sin\left[\frac{\phi_1 - \phi_2}{2}\right]\right]$
$A = 2\ A \sin\left(\frac{\phi_1 - \phi_2}{2}\right)$
$\frac{\phi_1 - \phi_2}{2} = \frac{\pi}{6}$
$\phi_1 = \frac{\pi}{3}$

@IITJEE_Advanced